

SESSION 04 DELIVERABLE

By: Margorie Fernanda Pérez Simba

1. Mini Systematic Review

a. Research Question

How can explainable, multimodal, and temporal artificial intelligence methods improve early student dropout prediction and support transparent retention decisions in higher education?

b. Review Method

i. Search Strategy

The search was run in Scopus on 30 September 2026, with the query applied to article titles, abstracts, and keywords to retrieve studies that connect artificial intelligence, student dropout or retention, explainability, and higher education. Scopus filters limited the results to English-language articles or conference papers from 2019 to 2026 at the final publication stage.

ii. Boolean Search String

("artificial intelligence" OR "machine learning" OR "deep learning") AND ("student dropout" OR "student attrition" OR "student retention") AND (explainab* OR interpretab* OR XAI) AND ("higher education" OR universit* OR college*)

iii. Eligibility Criteria

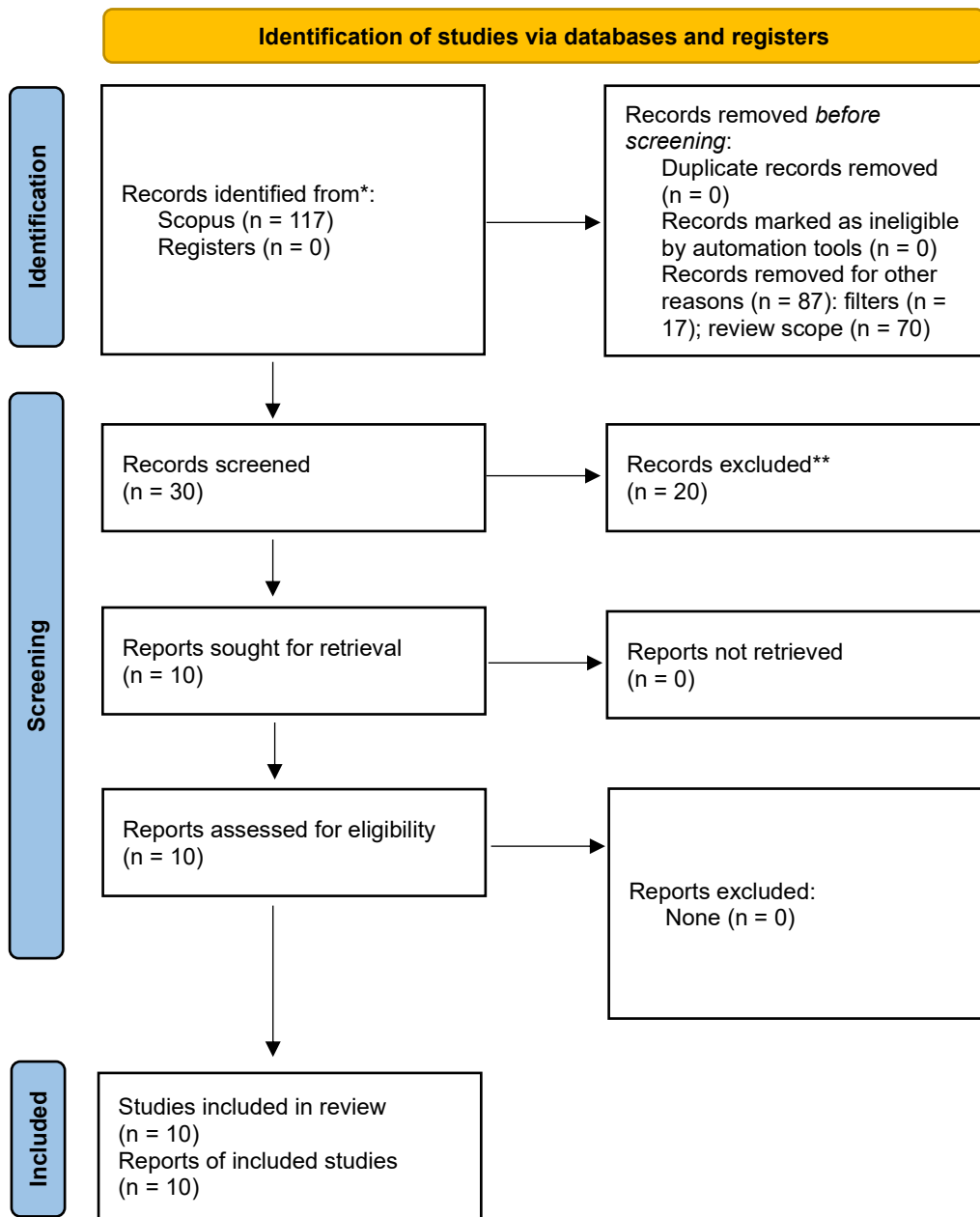
Criterion	Inclusion	Exclusion
Population and context	Students or student records in higher education.	K-12, vocational education, or contexts unrelated to higher education.
Phenomenon	Student dropout, attrition, retention, persistence, or academic outcome prediction directly connected to dropout risk.	Studies without a clear relationship to student dropout or retention.
Technology	Machine learning, deep learning, AutoML, or an AI-based prediction framework.	Purely descriptive studies without an AI or machine learning component.
Explainability	Explicit use or analysis of explainability, interpretability, XAI, SHAP, LIME, surrogate models, or transparent decision support.	Prediction studies that did not address interpretability or explainability.
Evidence	Full text available in English with sufficient methodological and bibliographic information.	Unavailable full text, insufficient methodological detail, or no traceable publication identifier.
Integrity	No author-related conflict in selecting the final evidence set.	An article authored by the researcher was excluded to reduce possible selection bias.

iv. Screening and Selection

The Scopus search returned 117 records. Applying the predefined year, language, document type, and publication-stage filters excluded 17 records, leaving 100. Because this was a classroom mini review, only the first 30 records ranked by relevance were screened; the other 70 were not assessed, a limitation of scope. Of the 30 screened by title and abstract, 20 were excluded for not meeting the topical, methodological, accessibility, or integrity criteria, including one article authored by the researcher (to avoid selection bias) and one candidate without a DOI or other persistent identifier. The 10 remaining reports were retrieved and met every criterion at the eligibility stage, so the final corpus contains 10 studies.

No.	Study	Context and data	Methods and XAI	Main evidence
1	Albugami et al. (2026)	Saudi universities; 4,560 questionnaire responses	12 ML models; SHAP, LIME, and LEAF	KNN with RFE reached 97.6% accuracy and F1 score; LIME exceeded SHAP on LEAF measures.
2	Yuan and Fan (2026)	Public dataset (not named; variables match the Portuguese UCI data); sample size not reported	LR, SVM, RF, XGBoost, Stacking; SHAP	Academic performance and tuition status were major predictors; dataset reporting was incomplete; the optimal model's confusion matrix is labeled SVM.
3	Lopez-Muñoz et al. (2026)	Systematic review; 75 studies from 2019 to 2024	Review of ML, DL, and XAI methods	About 68% were single-institution; static data, weak theory integration, limited fairness analysis, and little intervention validation (71%) were common.
4	Bettahi et al. (2025)	Moroccan institution; 17,883 records; replication on the Portuguese UCI dataset	Nine classifiers; SHAP	XGBoost reached AUC-ROC 0.993, F1 0.911, and recall 0.944; temporal dynamics were not modeled.
5	Kwon (2026)	South Korean university; 74,073 cleaned semester records	Rolling windows, CatBoost, XGBoost, Stacking; calibration and SHAP	The strongest temporal study; predictions covered one to three semesters, but external and intervention validation remain pending.
6	Padmasiri and Kasthuriarachchi (2024)	Kaggle dataset (matches Portuguese UCI data) and 15 academic professionals	Five classifiers; SHAP and LIME web interface	The interface supported decisions, but explanation clarity was moderate and the user sample was small.
7	Alcauter et al. (2023)	Private Mexican university; 4,709 engineering student records	DT, RF, SVM, ANN; LIME	RF reached 97.8% accuracy; the study was limited to one school and static modeling.
8	Umendu et al. (2026)	Public UCI (Portuguese) dataset; 4,424 student records	Extra Trees, RF, Gradient Boosting, LR; SHAP	Extra Trees reached AUC 0.96 and accuracy 87.4%; longitudinal and institutional validation were absent.
9	Radulescu and Balas (2026)	Bibliometric analysis of 352 documents and proof of concept with 4,424 students	VOSviewer and X-EWS framework; SHAP or LIME	The framework addresses explanation, fairness, monitoring, and intervention, but still requires real institutional validation.
10	Nawae et al. (2026)	Portuguese public dataset; 4,424 students	AutoGluon; surrogate tree and LIME	XGBoost reached approximately 0.899 across metrics; surrogate fidelity was approximately 0.77.

2. PRISMA 2020 Flow Diagram



* Records were identified from Scopus only; no other databases or registers were searched.

** All 20 records were excluded manually by the author; no automation tools were used at the screening stage.

Source: Page MJ, et al. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.

3. Gap Analysis Table

Gap type	Evidence	Identified gap	Response of the proposed research
Knowledge gap	Most studies report predictive performance and influential variables but do not test whether explanations improve decisions (Kwon, 2026; Radulescu & Balas, 2026). Lopez-Muñoz et al. (2026) found that 71% of the 75 studies reviewed lacked empirical validation of interventions. The only formal user evaluation among the included studies (15 academic professionals) rated explanation clarity as moderate (Padmasiri & Kasthuriarachchi, 2024).	It is still unclear whether explainable predictions are understandable, actionable, and useful for transparent retention decisions.	Evaluate predictive performance together with the clarity, usefulness, and decision-support value of the explanations.
Methodological gap	Static classification on institutional snapshots remains dominant (Lopez-Muñoz et al., 2026). Bettahi et al. (2025) did not model temporal dynamics, and Umendu et al. (2026) list longitudinal data as future work. Kwon (2026) models time but relies mainly on structured registry data; no study here analyzes text or platform logs.	Few approaches integrate multimodal institutional data, temporal risk evolution, and explainability in one validated framework.	Design an explainable, multimodal, and temporal AI framework integrating academic, administrative, textual, and period-based information.
Contextual gap	About 68% of the studies reviewed by Lopez-Muñoz et al. (2026) rely on a single institution. Six of the ten studies here use or replicate on the same Portuguese dataset, and the only Latin American study covers engineering students at one private Mexican university (Alcauter et al., 2023).	Few studies examine how institution-specific dropout definitions and retention processes shape prediction and explanation in Latin American higher education.	Develop and evaluate the framework in one higher education institution using its formal dropout definition and operational retention processes.
Theoretical gap	Lopez-Muñoz et al. (2026) found that only about 6% of the 75 studies reviewed turned persistence theory into features or model design. Umendu et al. (2026) are a partial exception, aligning predictors and SHAP results with Tinto's and Astin's theories.	Educational persistence theory and AI-based dropout prediction remain weakly connected.	Use concepts from student persistence and retention theory to guide predictor selection, interpretation, and intervention categories.

4. References

- Albugami, S., Wali, A., & Almagrabi, H. (2026). Predicting student dropout in Saudi universities using machine learning and explainable AI. *PeerJ Computer Science*, 12, e3490. <https://doi.org/10.7717/peerj-cs.3490>
- Alcauter, I., Martinez-Villaseñor, L., & Ponce, H. (2023). Explaining factors of student attrition at higher education. *Computación y Sistemas*, 27(4), 929–940. <https://doi.org/10.13053/CyS-27-4-4776>
- Bettahi, A., Belouadha, F. Z., & Harroud, H. (2025). A modular and explainable machine learning pipeline for student dropout prediction in higher education. *Algorithms*, 18(10), 662. <https://doi.org/10.3390/a18100662>
- Kwon, J. B. (2026). A portable, generalizable machine learning framework for long-term student dropout prediction. *IEEE Access*, 14, 30830–30843. <https://doi.org/10.1109/ACCESS.2026.3666870>
- Lopez-Muñoz, G. L., Gonzalez Serrano, C., & Sanchez-Ferreira, C. (2026). Student dropout prediction in higher education: A systematic review of machine learning methods and risk factors. *Journal of Information Technology Education: Research*, 25, Article 20. <https://doi.org/10.28945/5776>
- Nawae, M., Chankua, S., & Longsaman, M. (2026). Unpacking the black box: A hybrid XAI framework for AutoGluon-based multiclass student outcome prediction. *Journal on Efficiency and Responsibility in Education and Science*, 19(1), 28–39. <https://doi.org/10.7160/eriesj.2026.190103>
- Padmasiri, P., & Kasthuriarachchi, S. (2024). Interpretable prediction of student dropout using explainable AI models. In *2024 International Research Conference on Smart Computing and Systems Engineering (SCSE)* (pp. 1–7). IEEE. <https://doi.org/10.1109/SCSE61872.2024.10550525>
- Radulescu, D. A., & Balas, V. E. (2026). Explainable early warning systems for student dropout prediction: Insights from a bibliometric analysis. *International Journal of Computers Communications & Control*, 21(4), Article 7599. <https://doi.org/10.15837/ijccc.2026.4.7599>
- Umendu, E. C., Ghanzanfar, M., Kans, A., & Ahad, M. A. R. (2026). Enhancing student retention in higher education institutions: Machine learning approach. *Electronics*, 15(4), 734. <https://doi.org/10.3390/electronics15040734>
- Yuan, Y., & Fan, Z. (2026). Dropout risk in higher education via multidimensional features and machine learning. In *The 2026 International Conference on Artificial Intelligence in Education Technology and Data Science* (pp. 25–30). ACM. <https://doi.org/10.1145/3810158.3810167>